

## 1. General Description

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The EMS4567 is a high performance, quad, Single Pole Double Throw (SPDT) analog switch that features ultra-low  $R_{ON}$  of  $1.9\ \Omega$  (typical) at  $5.5V\ V_{CC}$ . The EMS4567 operates over a wide  $V_{CC}$  range of  $1.5V$  to  $5.5V$  and is designed for break-before-make operation. The select input is TTL-level compatible.

EMS4567 is also featured with smart circuitry to minimize  $V_{CC}$  leakage current even when the control voltage is lower than  $V_{CC}$  supply voltage. This feature suits mobile handset applications by allowing direct interface with baseband processor general-purpose IO with minimal battery consumption. In other word, there is no need of additional device to shift control level to be the same as that of  $V_{CC}$  in real application.

The EMS4567 is available in QFN1.8x2.6-16L package. Standard Products are Pb-free and halogen-free.

## 2. Features and Benefits

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- Wide supply voltage range from  $1.5\ V$  to  $5.5\ V$
- Low On-resistance,  $R_{ON} = 1.9\ \Omega$  when  $V_{CC} = 5.5\ V$
- High Bandwidth:  $-3\ dB$  @  $700\ MHz$
- Rail-to-Rail Signal Range
- Break-Before-Make Switching
- Low quiescent current over an Expanded Control Input Range
- ESD protection: HBM ESDA/JEDEC JS-001 Class 2 exceeds  $2000\ V$

## 3. Applications

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- Cell phones, PDA, Digital Camera and Notebook
- LCD Monitor, TV and Set-Top Box
- Audio and Video Signal Routing
- Other electronic equipment

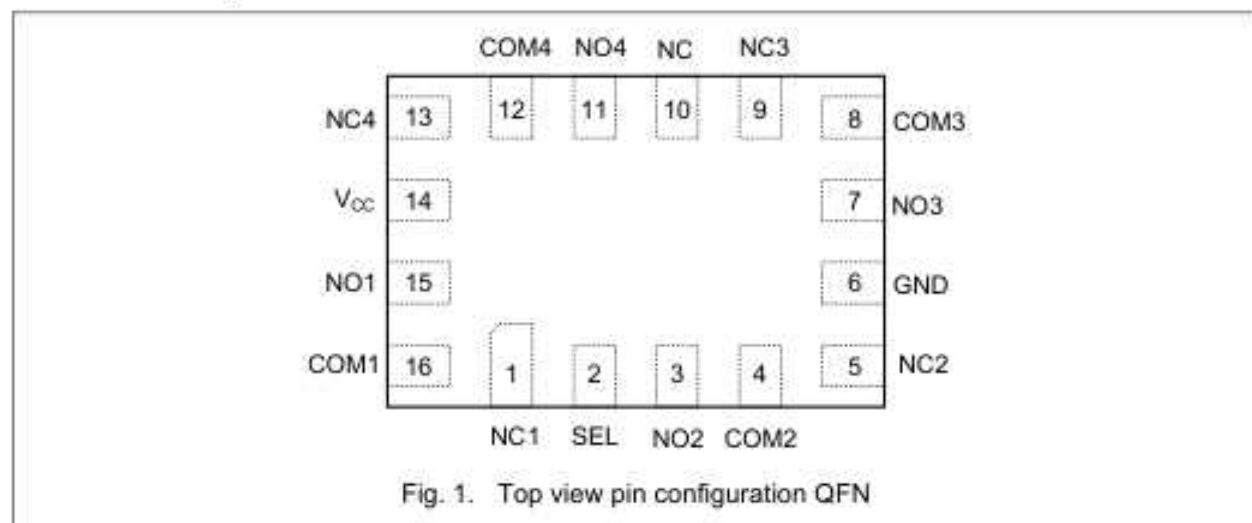
## 4. Ordering Information

**Table 1. Ordering information**

| Type number | Package        |                                                              |          |
|-------------|----------------|--------------------------------------------------------------|----------|
|             | Name           | Description                                                  | Quantity |
| EMS4567RSV  | QFN1.8x2.6-16L | QFN package, 16 pins<br>1.8 mm × 2.6 mm; 0.6 mm (Max) height | 3000     |

## 5. Pinning Information

### 5.1. Pinning



### 5.2. Pin description

**Table 2. Pin description**

| Symbol              | Pin       | Description                 |
|---------------------|-----------|-----------------------------|
| COM1,COM2,COM3,COM4 | 4,8,12,16 | Common Port                 |
| NC1,NC2,NC3,NC4     | 1,5,9,13  | Data Port (Normally closed) |
| NO1,NO2,NO3,NO4     | 15,3,7,11 | Data Port (Normally open)   |
| SEL                 | 2         | Logic Input Control         |
| V <sub>cc</sub>     | 14        | Positive Power Supply       |
| GND                 | 6         | Ground                      |
| NC                  | 10        | Not Connect                 |

## 6.Function Diagram

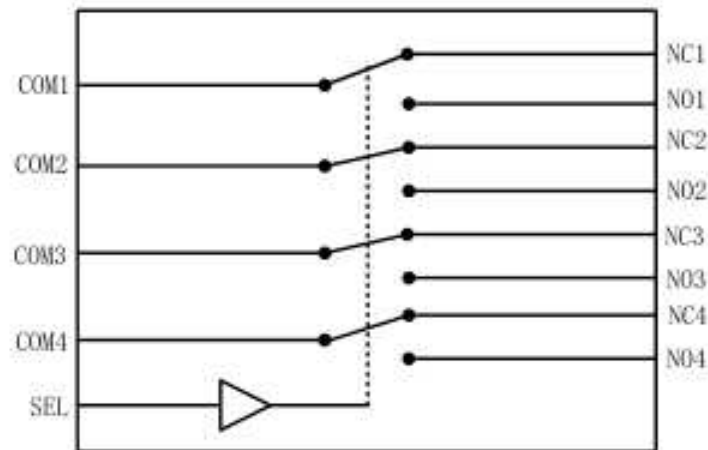


Fig. 2. Logic symbol

## 7.Function Description

**Table 3. Function table**

H = HIGH voltage level; L = LOW voltage level.

| SEL | Function                                                                                     |
|-----|----------------------------------------------------------------------------------------------|
| L   | NC1 Connected to COM1, NC2 Connected to COM2<br>NC3 Connected to COM3, NC4 Connected to COM4 |
| H   | NO1 Connected to COM1, NO2 Connected to COM2<br>NO3 Connected to COM3, NO4 Connected to COM4 |

## 8. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

**Table 4. Absolute Maximum Ratings**

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

| Symbol          | Parameter                                                          | Conditions | Min  | Max  | Unit |
|-----------------|--------------------------------------------------------------------|------------|------|------|------|
| $V_{CC}$        | supply voltage                                                     |            | -0.3 | 6.5  | V    |
| $V_{IN}$        | control Input Voltage                                              |            | -0.3 | 6.5  | V    |
| $V_A$           | signal input voltage                                               |            | -0.3 | 6.5  | V    |
| $I_A$           | continuous Current Through COM, NC, NO                             |            |      | ±100 | mA   |
|                 | peak Current Through COM, NC, NO<br>(pulsed at 1ms 50% duty cycle) |            |      | ±200 | mA   |
| $T_J$           | junction Temperature under Bias                                    |            |      | 150  | °C   |
| $T_L$           | lead Temperature (Soldering, 10 seconds)                           |            |      | 260  | °C   |
| $R_{\theta JA}$ | thermal resistance                                                 |            |      | 350  | °C/W |

## 9. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

**Table 5. Recommended Operating Conditions**

| Symbol    | Parameter             | Conditions | Min | Max      | Unit |
|-----------|-----------------------|------------|-----|----------|------|
| $V_{CC}$  | supply voltage        |            | 1.5 | 5.5      | V    |
| $V_{IN}$  | control input voltage |            | 0   | 5.5      | V    |
| $V_A$     | signal input voltage  |            | 0   | $V_{CC}$ | V    |
| $T_{amb}$ | ambient temperature   |            | -40 | 125      | °C   |

## 10. Electrical Characteristics

**Table 6. Static characteristics**
 $T_{amb}=25\text{ }^{\circ}\text{C}$ ,  $V_{CC}=3.3\text{ V}$ , unless otherwise specified.

| Symbol                               | Parameter                                               | Conditions                                                                               | Min | Typ  | Max       | Unit          |
|--------------------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------|-----|------|-----------|---------------|
| <b>DC Electrical Characteristics</b> |                                                         |                                                                                          |     |      |           |               |
| $V_{IH}$                             | input logic high level                                  | $V_{CC} = 3.4\text{ V to }5.5\text{ V}$                                                  | 1.6 |      |           | V             |
|                                      |                                                         | $V_{CC} = 1.5\text{ V to }3.3\text{ V}$                                                  | 1.4 |      |           | V             |
| $V_{IL}$                             | input logic low level                                   | $V_{CC} = 3.4\text{ V to }5.5\text{ V}$                                                  |     |      | 0.6       | V             |
|                                      |                                                         | $V_{CC} = 1.5\text{ V to }3.3\text{ V}$                                                  |     |      | 0.4       | V             |
| $I_{CC}$                             | supply current                                          | $I_A = 0$ , $V_{SEL} = 0$ or $V_{SEL} = V_{CC}$                                          |     |      | 1.0       | $\mu\text{A}$ |
| $\Delta I_{CC}$                      | additional supply current                               | $I_A = 0$ , $V_{CC} = 2.5\text{ V or }4.2\text{ V}$<br>$V_{SEL} = 1.6\text{ V}$          |     |      | 3.0       | $\mu\text{A}$ |
| $I_{off}$                            | OFF-state switch leakage current from COM to NC (or NO) | $V_A = 5\text{ V}$ , $V_{NC}$ or $V_{NO} = 0\text{ V}$                                   |     |      | $\pm 2.0$ | $\mu\text{A}$ |
| $R_{ON}$                             | on-resistance                                           | $V_{CC} = 3.5\text{ V}$ , $V_A = 0.8\text{ V to }2.0\text{ V}$ ,<br>$I_A = 10\text{ mA}$ |     | 3.4  | 4.0       | $\Omega$      |
|                                      |                                                         | $V_{CC} = 4.5\text{ V}$ , $V_A = 0.8\text{ V to }2.3\text{ V}$ ,<br>$I_A = 10\text{ mA}$ |     | 2.4  | 3.0       | $\Omega$      |
|                                      |                                                         | $V_{CC} = 5.5\text{ V}$ , $V_A = 0.8\text{ V to }2.6\text{ V}$ ,<br>$I_A = 10\text{ mA}$ |     | 1.9  | 2.5       | $\Omega$      |
| $R_{FLAT}$                           | on-resistance flatness                                  | $V_{CC} = 3.5\text{ V}$ , $V_A = 0.8\text{ V to }2.0\text{ V}$ ,<br>$I_A = 10\text{ mA}$ |     | 0.2  | 0.5       | $\Omega$      |
|                                      |                                                         | $V_{CC} = 4.5\text{ V}$ , $V_A = 0.8\text{ V to }2.3\text{ V}$ ,<br>$I_A = 10\text{ mA}$ |     | 0.25 | 0.5       | $\Omega$      |
|                                      |                                                         | $V_{CC} = 5.5\text{ V}$ , $V_A = 0.8\text{ V to }2.6\text{ V}$ ,<br>$I_A = 10\text{ mA}$ |     | 0.3  | 0.5       | $\Omega$      |
| $\Delta R_{ON}$                      | on-resistance matching between channels                 | $V_{CC} = 3.5\text{ V}$ , $V_A = 0.8\text{ V to }2.0\text{ V}$ ,<br>$I_A = 10\text{ mA}$ |     | 0.1  | 0.2       | $\Omega$      |
|                                      |                                                         | $V_{CC} = 4.5\text{ V}$ , $V_A = 0.8\text{ V to }2.3\text{ V}$ ,<br>$I_A = 10\text{ mA}$ |     | 0.1  | 0.2       | $\Omega$      |
|                                      |                                                         | $V_{CC} = 5.5\text{ V}$ , $V_A = 0.8\text{ V to }2.6\text{ V}$ ,<br>$I_A = 10\text{ mA}$ |     | 0.1  | 0.2       | $\Omega$      |

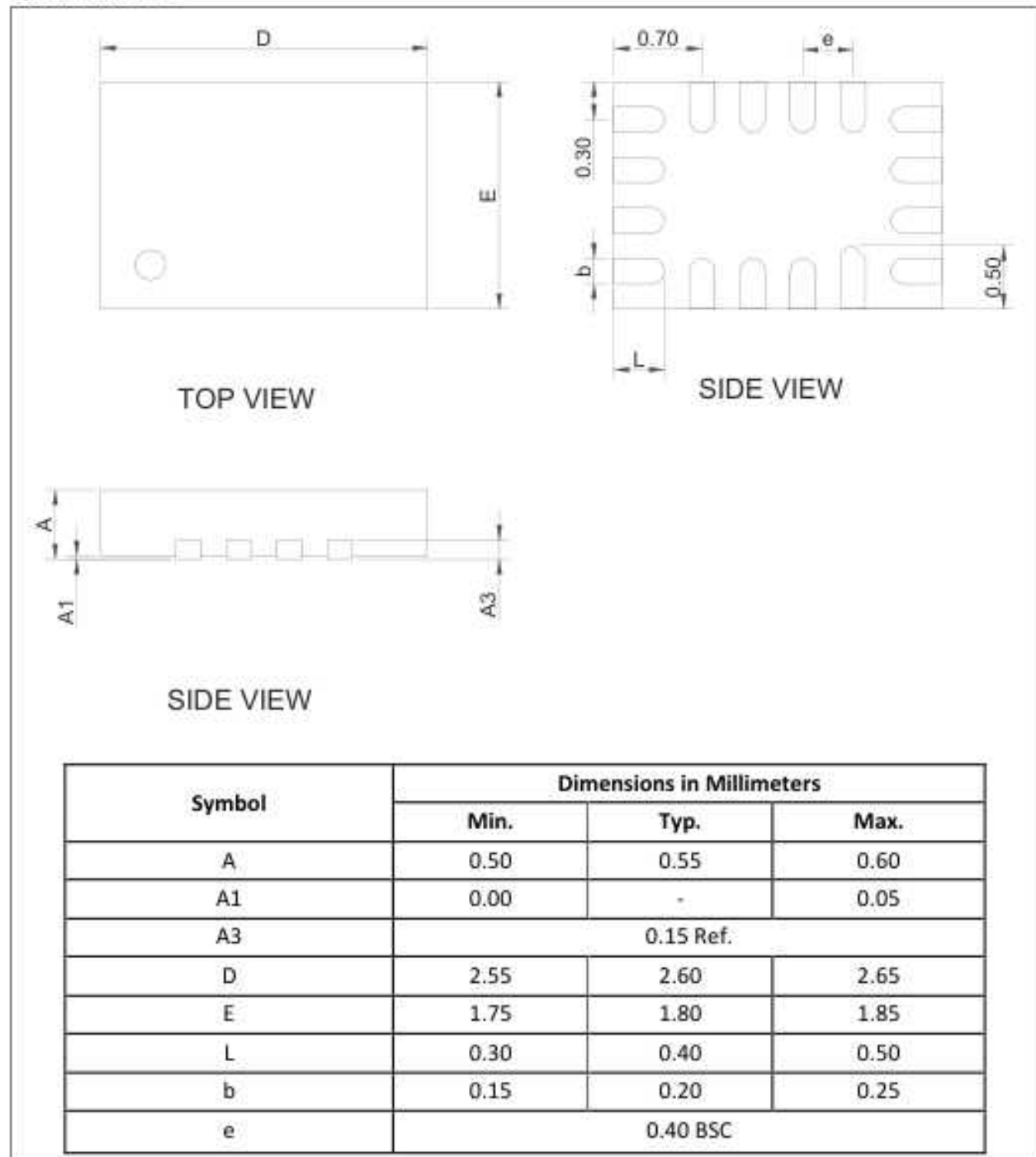
## EMS4567

### Low On Resistance, Quad SPDT Analog Switch

| Symbol                                | Parameter                 | Conditions                                                            | Min | Typ | Max | Unit |
|---------------------------------------|---------------------------|-----------------------------------------------------------------------|-----|-----|-----|------|
| <b>AC Electronics Characteristics</b> |                           |                                                                       |     |     |     |      |
| $t_{ON}$                              | turn-On Time              | $V_A=1.5V$ , $C_L=50pF$ , $R_L=50\Omega$                              |     | 470 |     | ns   |
| $t_{OFF}$                             | turn-Off Time             | $V_A=1.5V$ , $C_L=50pF$ , $R_L=50\Omega$                              |     | 120 |     | ns   |
| $t_{B8M}$                             | break-before-make time    | $V_A=1.5V$ , $C_L=50pF$ , $R_L=50\Omega$                              |     | 600 |     | ns   |
| BW                                    | -3dB Bandwidth            | $C_L=0pF$                                                             |     | 800 |     | MHz  |
| OIRR                                  | off isolation             | $f=10\text{ KHz}$                                                     |     | -80 |     | dB   |
|                                       |                           | $f=240\text{ MHz}$                                                    |     | -31 |     | dB   |
| $X_{TALK}$                            | crosstalk                 | $f=10\text{ KHz}$                                                     |     | -82 |     | dB   |
|                                       |                           | $f=240\text{ MHz}$                                                    |     | -26 |     | dB   |
| THD                                   | total harmonic distortion | $f=20\text{Hz to } 20\text{KHz}$<br>$V_A=600mV_{pp}$ , $R_L=32\Omega$ |     | -80 |     | dB   |
| <b>Capacitance</b>                    |                           |                                                                       |     |     |     |      |
| $C_{ON}$                              | On Capacitance            | $f=100\text{KHz}$                                                     |     | 12  |     | pF   |
| $C_{OFF}$                             | Off Capacitance           | $f=100\text{KHz}$                                                     |     | 4   |     | pF   |

## 11. Package Outline

QFN1.8x2.6-16L



## 12. Revision History

Table . Revision history

| Document ID      | Release Date | Data sheet status | Change notice | Supersedes |
|------------------|--------------|-------------------|---------------|------------|
| EMS4567 Rev. 1.0 | Oct 30, 2024 | Product datasheet |               |            |